Very little data tampering is needed to produce statistically significant results: A Monte Carlo simulation

Ian Hussey[a,*]

[a] *University of Bern,*

Abstract

Simulation studies to date have examined the impact of Questionable Research Practices on the false positive rate of common statistical inference tests (e.g., Stefan & Schonbrödt, 2023), but not the impact of unquestionably undesirable research practices, such as fabrication and falsification. While any result can be obtained through sufficient fabrication or falsification, we have little insight into how much of either is needed to generate statistically significant results under a null hypothesis. This article therefore simulates data generated under the true null hypothesis and alters, deletes, or invents one data point at a time, targeting the data points that most oppose the desired result, and re-running the test after every change until it is significant in the intended direction. Six tampering strategies (dropping cases, switching condition labels, re-pairing values, duplicating, fabricating, and replacing cases) are applied to three common tests: the independent t-test, Pearson r correlation, and the Chi-square test for a 2x2 contingency table. Results demonstrate that, in small samples, only a small absolute number of tampers are typically needed to obtain significant results. In large samples, although the absolute number of tampers needed grows, the proportion of the sample needing to be tampered with to obtain significance falls rapidly. The strategies that alter no recorded value, and so are plausibly hardest to detect, are among the most effective. Critically, unjustifiably deleting inconvenient data points, which researchers rate as far more defensible than invention, is no less dangerous than outright fabrication: both manufacture false positives from null data reliably, and at comparable and very small cost (e.g., median tampers of 5-10% at moderate sample sizes across tests). Fabricating and falsifying results therefore requires a worryingly small amount of actual data tampering.

1. Introduction

In November 2019, the pivotal trial of a treatment for ANCA-associated vasculitis was unblinded and its primary analysis run. The result was not significant ($p = .1025$). According to a subsequent letter from the US Food and Drug Administration, unblinded personnel employed by the sponsor then selected nine of the trial's 331 patients for re-adjudication of their remission status, having first confirmed that reclassifying five of

*Corresponding author
Email address: `ian.hussey@unibe.ch` (Ian Hussey)

them would move the study across the significance threshold. The five were reclassified, the database was re-locked, and the re-run analysis returned $p = .0132$. That analysis was the only one submitted in support of approval, which was granted in 2021. In April 2026 the FDA proposed to withdraw it, concluding that the integrity of the trial had been compromised; the sponsor contests this and a hearing is pending (U.S. Food and Drug Administration, Center for Drug Evaluation and Research, 2026). Whatever its resolution, one feature of the case is worth dwelling on. Five records out of 331, just 1.5% of the sample, separated a failed trial from a successful one.

How often does it occur that data are simply tampered with until statistical significance is reached? Meta-analysis of the survey literature puts self-admitted research misconduct at 2.9% of researchers (95% CI [2.1, 3.8]), comprising 1.9% for fabrication and 3.3% for falsification, while 15.5% report having witnessed it in others (Xie et al., 2021; see also Fanelli, 2009). When the National Survey on Research Integrity used a randomized response design to give respondents provable deniability, admitted fabrication rose to 4.3% and falsification to 4.2% (Gopalakrishna et al., 2022). From the other side of the desk, 24% of consulting biostatisticians in a US national survey had been asked, at least once in the preceding five years, to remove or alter data records to better support a research hypothesis (Wang et al., 2018). These are not the rates of a vanishingly rare phenomenon, and each is likely a lower bound of the true prevalence.

Prevalence, though, is only half the question. The other half is how 'effective' such data manipulations actually are. Here, the literature is largely silent. Chambers (2017) set out a "dirty dozen" of ways to guarantee significant results even under the null hypothesis, the second of which is switching participants between conditions. Brown (2020) posted a short simulation of the efficacy of this in a blog post. Both convey the same intuition that a handful of unjustified changes to the data is enough to produce statistically significant results in many cases, but neither is a systematic assessment. The field has no systematic account of how much tampering a given strategy requires, whether that requirement grows with sample size, or how the strategies compare with one another.

For the ambiguous end of the continuum of merely 'questionable' research practices, that gap has been filled. Stefan & Schönbrodt (2023) compiled 12 p-hacking strategies from the literature, identified the factors controlling each one's severity, and simulated the resulting increase in false-positive rates. Their scope, as they state it, is the "grey area between good practice and outright scientific misconduct". They stop deliberately at its far edge: discussing misreported test statistics, they note that at some point one must ask whether a practice is still questionable or has become outright fraud. As such, the literature says comparably little about the impact of indefensible research practices on false-positive rates. Of course, given a willingness to engage in an arbitrary amount of data falsification or fabrication, by definition one can obtain statistically significant results. Here, I ask a narrower question: how much data tampering, of what form, is typically needed to acquire significant results.

Quantifying the indefensible practices is worth doing because of how researchers appear to decide what to do. Entradas et al. (2026) find that perceived seriousness is by far the strongest predictor of whether a researcher engages in a questionable practice, ahead of every career and demographic factor they measure. Equally, in John et al. (2012), deciding whether to exclude data after seeing its effect on the results was admitted by

38–43% of psychologists and rated 1.61 out of 2 for defensibility, while falsifying data was admitted by 0.6–1.7% and rated 0.16. These results are consistent with the idea of perceived norms shaping research behaviors. That is, if researchers are more likely to engage in practices they understand to be seriously problematic and are less likely to engage in ones that are seen as such. However, the boundary between these plausibly varies between individuals, research communities, and over time, creating what is effectively an Overton Window of questionable versus unacceptable research practices. Many research practices that were regarded as acceptable prior to the invention of the term *p*-hacking are now considered in many research communities to be substantially less acceptable. If perceptions of how problematic a given set of research behaviors are influence the prevalence of those behaviours, then establishing how relatively bad a given set of practices actually are, how little effort they take, and how much bias they inject, likely represents important information to dissuading their practice.

In this research, I explicitly adopt a social-epistemological approach, which understands such researcher behaviors as a set of evolving community norms rather than (im)moral acts. Equally, this perspective is concerned only with the undesirable statistical consequences of certain researcher behaviors, but is agnostic to researchers' intent when taking them. With that being said, at a human level, it is easier to imagine a researcher who deletes a small number of data points to 'reveal' an effect they strongly believe is there, lying just under the surface of the data, when it is not (e.g., cases of 'qualitative exclusion strategies': Schiller et al. (2018)) than to imagine scientists simply making up datasets wholesale [although the latter does occur; Verfaellie & McGwin (2011)], despite both strategies producing false-positive results. Regardless of whether these two scenarios can be equated in their intent, it is an open question whether they can be equated in their outcomes, as it is generally unclear how relatively robust or fragile most results are to modifications of only a small number of data points (although see Spiess et al., 2024).

Some definitional matters are worth considering, insofar as they informed which tampering strategies were simulated. The US government's Office of Research Integrity provides influential definitions of research misconduct and its forms (Steneck, 2007). According to the ORI definitions, fabrication is "making up data or results and recording or reporting them". This is represented by the fabrication, duplication and replacement strategies simulated here, which add data points. Falsification is "manipulating research materials, equipment, or processes, or changing or omitting data or results such that the research is not accurately represented in the research record". This is represented by the dropping strategy (which omits observations) and the condition switching and re-pairing strategies (which change labels and pairings without adding or deleting anything). The taxonomy that carries the regulatory and moral weight thus classifies by mechanism: was the data point invented, or was it altered or omitted? Of course, the line is not necessarily clear here, given that an altered data point is equivalent to a removed data point plus a newly added data point.

In this article I do three things. First, I formalize data tampering as a family of iterative algorithms (dropping, condition switching, re-pairing, duplication, fabrication, and replacement) applied to three of the most common elementary designs: the two-group mean difference (Student's *t*-test), the Pearson correlation, and the 2×2 association (chi-square test). One caveat attaches to the strategies that invent values (fabrication and replacement): once an analyst is willing to make up data, the invented values can be of any extremity, so any particular choice of fabricated values is one point in a large space of experimenter degrees of freedom trading efficiency per fabricated point against plausibility. The fabricated values simulated here are deliberately aggressive, and the relative cost of these strategies is therefore harder to benchmark than that of strategies that only delete or relabel existing data. Second, I estimate the effort each strategy requires, with the full design specified under the ADEMP framework (Morris et al., 2019; Siepe et al., 2024) and every estimate accompanied by a Monte Carlo standard error. I invert the design used in previous simulations to understand the impact of p-hacking. Where Stefan & Schönbrodt (2023) fix a budget of manipulations and measure the resulting false-positive rate, I fix the goal (statistical significance) and measure the number of tampered data points required to reach it. I take this to be the more ecologically valid framing for deliberate tampering: an analyst who has decided to manufacture a result does not set a budget in advance and stop when it is exhausted, but continues until the p-value crosses. It also yields effort directly, in units of edited data points. Because the simulations run without any tampering budget, the complete distribution of required effort is observed, and the proportion of datasets exceeding any given budget is recovered afterwards from the same runs. Third, I show that the effort required is small, that it does not grow with sample size in proportional terms, and that the strategies that are likely hardest to detect are among the cheapest.

2. Methods

The design is specified under the ADEMP framework (Aims, Data-generating processes, Estimands, Methods, Performance measures) (Morris et al., 2019), as operationalized for psychology by Siepe et al. (2024). The three research compendia containing all simulation code, per-strategy results, and extended rationale are available in the project repository (see *Data and code availability*); this section compresses them.

2.1. Aims

The study is descriptive: there is no trivial, well-established closed-form benchmark for the minimum number of tampered points needed to manufacture significance — that number is a property of a tampering algorithm applied to random data, not an estimator of a population parameter — and I did not attempt to derive one. For each strategy and sample size I aim to estimate (1) how many tampered data points are required to first reach $p < .05$, absolutely and as a proportion of the sample, and how this scales with sample size; (2) the proportion of datasets requiring more than a given share of the sample (post hoc budgets of 10% and 25%), and the proportion for which significance is never attained at all; and (3) how the strategies compare on these measures.

2.2. Data-generating processes

All data are generated under the true null, so every significant result is a false positive produced entirely by the tampering.

- Mean difference. Two independent groups of `n_per_condition` observations each, $Y_{ij} \sim \mathcal{N}(\mu_j, \sigma^2)$ with $\mu_{\text{control}} = \mu_{\text{intervention}} = 0$ and $\sigma = 1$.
- Correlation. A bivariate Normal sample of n observations with $\rho = 0$ and unit variances.

- 2×2 association. Two groups (exposed, not exposed) of `n_per_condition` observations with a binary outcome drawn at the same Bernoulli prevalence in both groups; prevalence is varied (.20 vs .05) to contrast a common with a rare outcome.

Per-group sample sizes are 10 and 25 to 200 in steps of 25 (total N = 20–400; the correlation setting uses the same total Ns). Fully crossing strategy with sample size (and prevalence) gives 45 conditions in each continuous setting and 54 in the binary setting, each replicated 10,000 times.

2.3. Estimands

The estimand is `tampers_needed`: the number of tampered data points applied before the test first returns a significant result in the intended direction, reported both as a count and as a proportion of the sample. The compendia also record the effect size of each final tampered dataset, but those are not analysed here: this article is concerned with the effort required to manufacture a false positive rather than with the size of the effect that results.

Counting effort requires a convention for what one "tampered data point" is, and cross-strategy comparisons depend partly on that convention. The convention used throughout is the number of case records created, deleted, or altered by an operation: dropping a case counts one; switching one case's condition label counts one; a re-pairing swap counts two, because two cases' y-values change; fabricating a pair of cases counts two; replacing one case's value counts one. Alternative accountings exist — counting operations performed (under which a re-pairing swap would count one) or cells of the data matrix edited (under which a dropped case in the correlation setting would count two, its x and its y) — and the relative ordering of strategies, particularly between label-based and value-based operations, shifts somewhat between them. The per-case convention is used because a case is the unit an analyst edits, but comparisons between strategies whose operations differ in kind should be read with the convention in mind.

2.4. Methods (tampering strategies)

Every strategy is an iterative procedure: generate a null dataset, test it, and while the desired result has not been reached, apply one tampering operation targeting data points opposed to it, then re-test. Table 1 defines the operations. All strategies push toward a positive effect by convention; tests are two-sided at $\alpha = .05$ (equal-variance t-test; Pearson correlation test; chi-square test without continuity correction). Success is sign-consistent: an iteration counts as significant only when $p < .05$ *and* the manufactured effect is in the intended direction. A two-sided test is also significant, in roughly 2.5% of null datasets, with the effect in the *wrong* direction; such datasets are not counted as zero-effort successes, and wrong-direction significance encountered in passing during tampering does not stop the procedure.

Table 1: The tampering strategies and their operationalization in each setting. Each operation counts as one tampered data point. For binary data, duplication, fabrication, and replacement collapse onto the same atomic operation (adding a favourable cell) and are treated as a single strategy.

Strategy	Mean difference	Correlation	2×2 association
Dropping	Delete the most hostile case (lowest in intervention or highest in control)	Delete the most discordant case (most negative centred cross-product)	Delete a 1 from not-exposed or a 0 from exposed
Label switching	Move the most hostile case's *condition label* to the other group, alternating direction to keep groups balanced	*Re-pairing:* swap the y-values of two cases so the most discordant case receives the y matching its x-rank	Move a not-exposed 1 to exposed or an exposed 0 to not-exposed, alternating direction
Duplication	Duplicate a case from each group's most favourable 10%	Duplicate one of the most concordant cases	(coincides with fabrication)
Fabrication	Add invented cases: control $\sim \mathcal{N}(0, 0.2)$, intervention $\sim \mathcal{N}(5, 0.2)$	Add a high-leverage concordant pair at $\pm(3, 3)$ with small scatter	Add a 1 to exposed or a 0 to not-exposed
Replacement	Replace the most hostile case with an invented one (whichever of the two candidate replacements most lowers p)	Replace the most discordant case's y with a concordant high-leverage value	(coincides with fabrication)

The strategies differ in how the targeted data point is chosen, and only replacement is greedy in the strict sense of optimizing the p-value at each step: it tries both candidate replacements and keeps whichever most lowers p. Dropping and re-pairing target the most extreme or discordant case, a heuristic that usually but not always coincides with the p-optimal move; condition switching alternates direction to keep group sizes balanced rather than choosing the better of the two available moves; and duplication samples at random from the most favourable decile. The effort estimates therefore characterize these specific, deliberately simple procedures: a more calculating tamperer could do somewhat better and a clumsier one worse, so the estimates should be read as typical costs of naive-but-directed tampering rather than as minima.

The label-switching row deserves emphasis because of its forensic properties: condition switching alters no recorded value, only group labels, and re-pairing preserves both marginal distributions *exactly* (the set of y-values is only permuted), so neither leaves a trace in the distribution of any single variable. Swapping *entire* conditions, as opposed to individual cases, is handled analytically rather than simulated: it merely doubles the directional false-positive rate to α, adding $\frac{1}{2}\alpha$.

No tampering budget is imposed. Each strategy runs until a significant result in the intended direction is reached or a structural limit is hit, such as removing all data under dropping. A loop guard is included at 10× the sample size as a termination backstop, needed because one-step tampering can stall — most notably when a dataset shows a strong effect in the *opposite* direction, which must be traversed through the two-sided test's $p \approx 1$ valley. Iterations stopped by the guard are flagged separately from structural limits in the recorded results and are recorded as never attaining significance. They are rare and confined to the smallest samples: only 4 of the 144 conditions contain any, all at $N \leq 50$, with a maximum of 6.85% of runs in a single condition. Stalling can also end an iteration without the guard, when a strategy runs out of improving moves; this is the more common outcome and is reported with the results. Because the full unbudgeted counts are recorded, the proportion of datasets exceeding *any* tampering budget is computable after the fact; I report budgets of 10% and 25% of the sample.

2.5. Performance measures and Monte Carlo uncertainty

An iteration that never attains a significant result in the intended direction (hereafter, never attains significance) has no finite tamper count (i.e., is right-censored). All percentile-based summaries (median, 95% prediction interval) are therefore computed over all iterations with the never-significant ones ranked above every finite count; a percentile falling among them is reported as *never* rather than being silently replaced by the survivors' percentile, which would answer a different question ("given that tampering worked, how many edits?") and bias the effort downward. The mean cannot be computed this way at all (the censored counts are unbounded above), so means are reported only as conditional-on-success skew diagnostics. Two intervals are used. The interquartile range is plotted, because one condition (correlation duplication at the smallest sample size) has a 97.5th percentile at nearly nine times the sample and would otherwise set the scale for every panel. The 95% prediction interval is reported in Table 2. Effort as a proportion is expressed relative to the *original* sample size, so strategies that append cases, or re-tamper a case already tampered with, can exceed 1.

Every reported estimate carries a Monte Carlo standard error: binomial for proportions, closed-form for means, and nonparametric bootstrap for medians and for the bounds of the 95% prediction intervals (Morris et al., 2019; Siepe et al., 2024). The number of replications, K = 10,000 per condition, was chosen for the Monte Carlo precision it yields rather than by convention: it bounds the MCSE of any reported proportion at $\sqrt{.5 \times .5/10000} = .005$, and the resulting bootstrap MCSEs of the median tamper counts are small fractions of one data point (Table 2). In tables, decimal places are set from the MCSE (two significant figures, with the estimate rounded to match) rather than fixed per table. Simulations were parallelized with reproducible L'Ecuyer–CMRG streams, so results are identical regardless of worker count.

3. Results

3.1. Effort is small, and falls as a share of the sample as studies grow

Across both continuous settings the absolute number of tampered data points needed to manufacture significance grows with sample size, but only slowly: median counts run from 2–10 points at $N = 20$ to 5–26 at $N = 400$, a twentyfold increase in sample size met by only a 2.0- to 3.2-fold increase in effort (Figure 1, Figure 2). Expressed as a share of the sample, effort therefore falls: from 10.0%–50.0% at $N = 20$ to 1.2%–6.5% at $N = 400$, a reduction of between 6.2- and 10.0-fold depending on the strategy. Manufacturing a false positive is not merely no harder in a large study than in a small one; proportionally it is considerably easier. The intuition that a large sample is robust to a few altered observations is exactly backwards. Framed as a budget: at $N = 400$, the proportion of

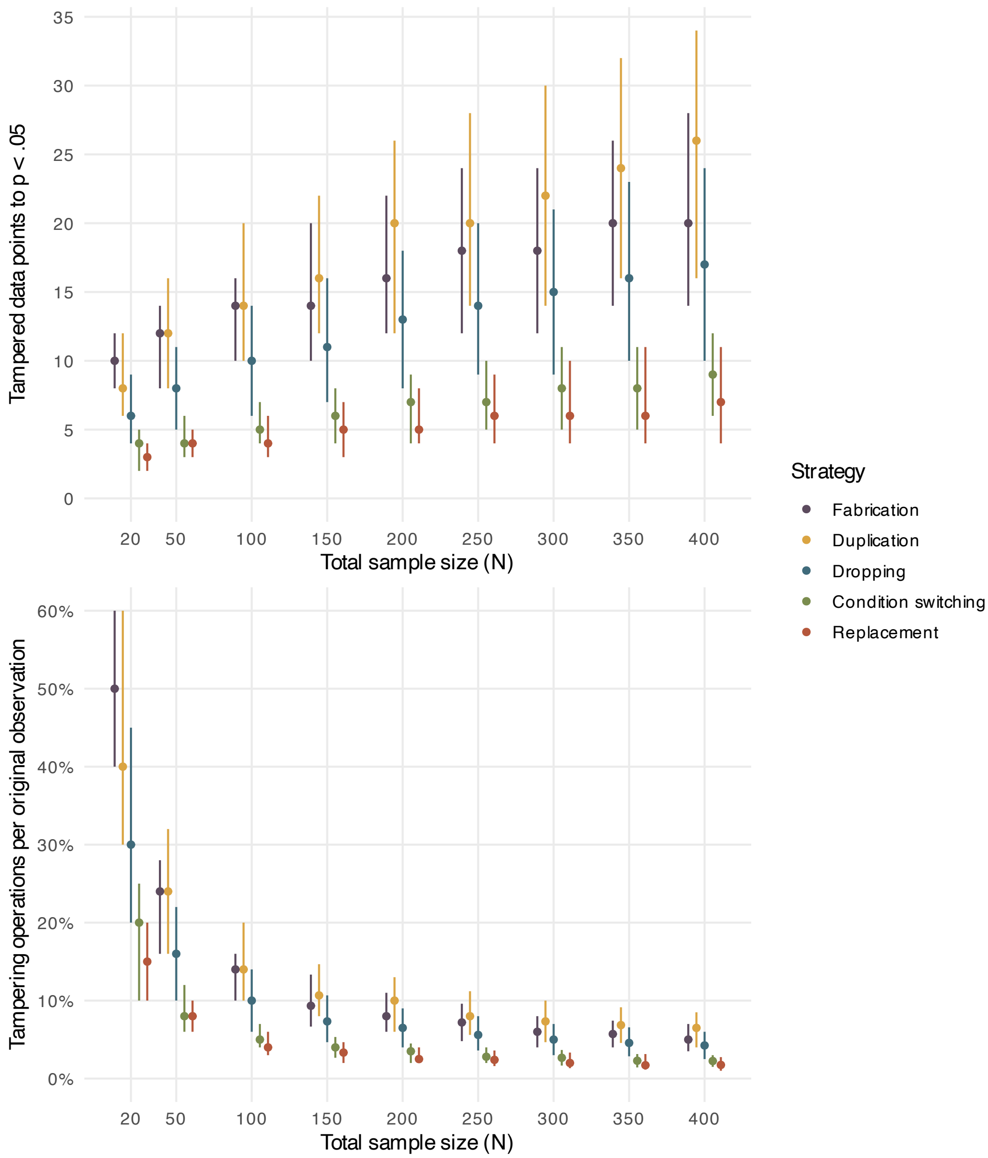

Figure 1: Mean-difference (t-test) setting: the number of tampered data points needed to first reach p < .05 (top) and the same effort as a proportion of the sample (bottom). Points are censoring-aware medians and bars are interquartile ranges. A bar ending in a hollow triangle is open-ended: more than a quarter of those datasets never attained significance, so the upper quartile has no finite value and the bar is drawn to the top of the data range. The lower panel expresses effort relative to the *original* sample size, so strategies that append cases or re-tamper the same case can exceed 1. Strategies are ordered by median proportion tampered, from the most expensive on the left to the cheapest on the right. K = 10,000 iterations per condition.

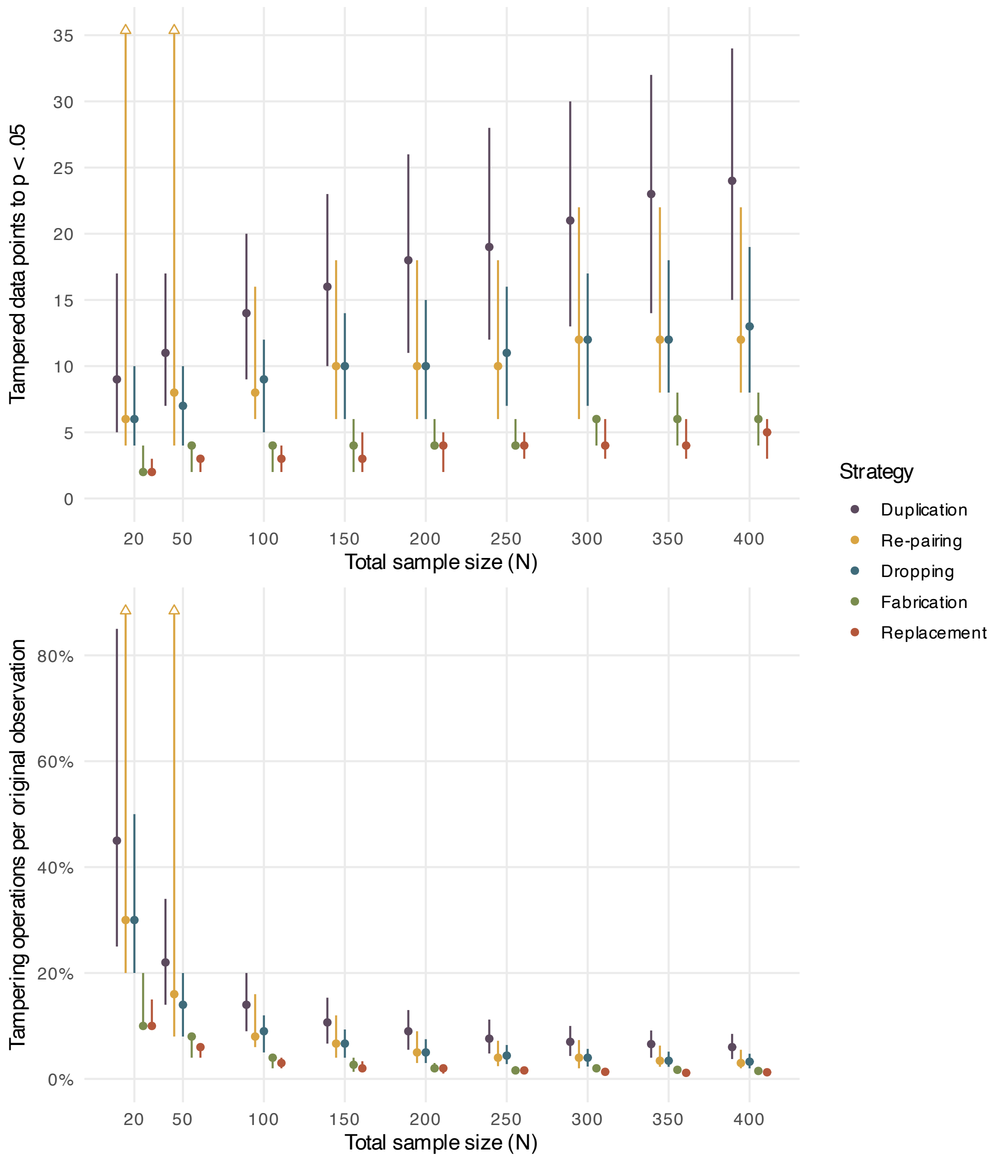

Figure 2: Correlation setting: tampered data points needed to first reach p < .05 (top) and as a proportion of the sample (bottom). Points are censoring-aware medians and bars are interquartile ranges; a bar ending in a hollow triangle is open-ended, its upper quartile falling among datasets that never attained significance. Effort in the lower panel is relative to the *original* sample size, so strategies that append cases can exceed 1. K = 10,000 iterations per condition.

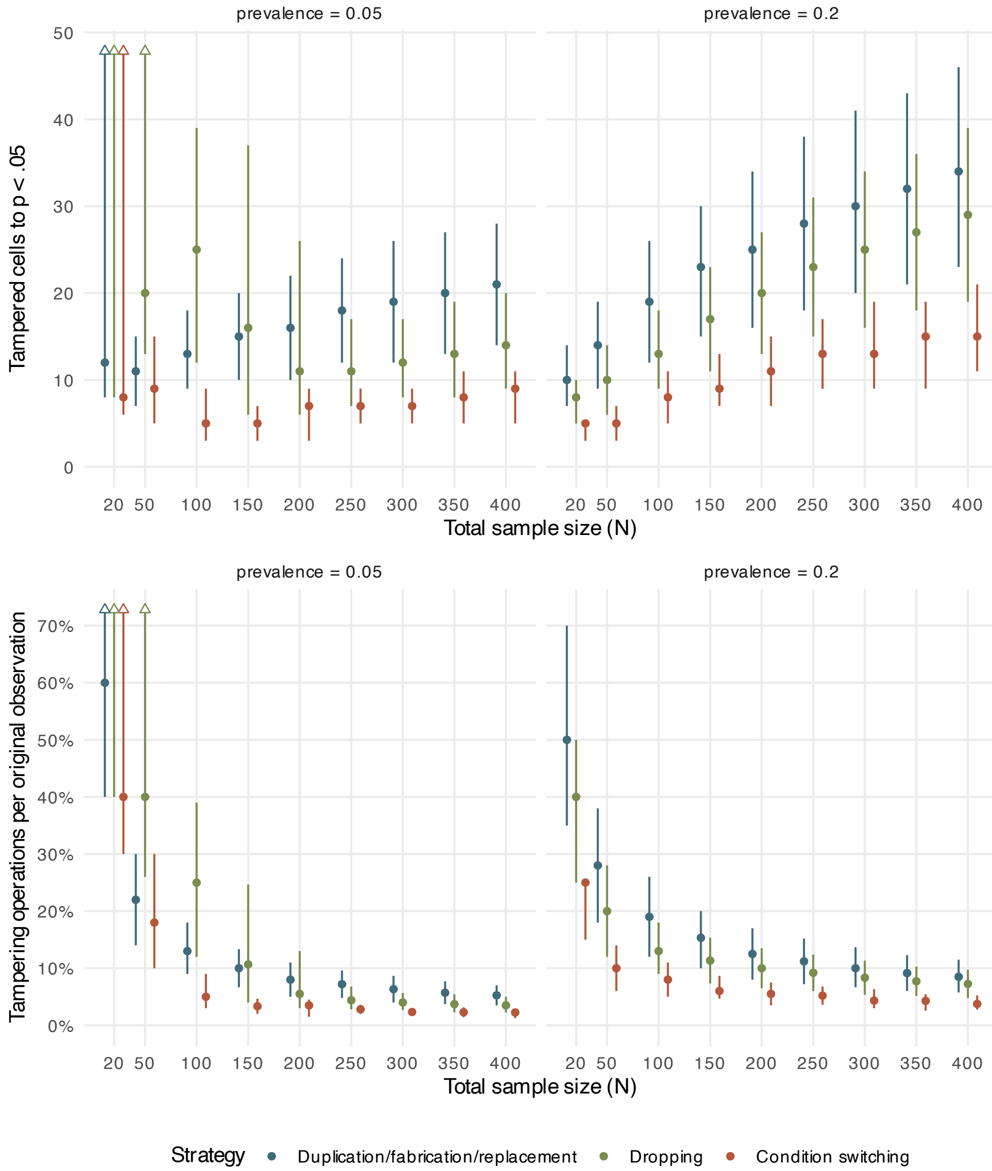

Figure 3: 2x2 association (chi-square) setting, by outcome prevalence: the number of tampered cells needed to first reach $p < .05$ (top) and the same effort as a proportion of the sample (bottom). Points are censoring-aware medians and bars are interquartile ranges; a bar ending in a hollow triangle is open-ended, its upper quartile falling among datasets that never attained significance. At prevalence .05 and N = 20, a majority of datasets cannot be pushed to significance at all by dropping, so even the median is not attained and no point is drawn for it. K = 10,000 iterations per condition.

datasets that would require more than 10% of the sample to be tampered is at most 22.2% for the least efficient continuous-setting strategy, and for the cheapest it is 0.0%. That worst case is re-pairing, and it is not a matter of expense: 22.2% of those datasets are ones the strategy fails on entirely, for reasons taken up below. Every other continuous strategy reaches significance in essentially every dataset at that sample size.

The 2×2 setting is the partial exception that illustrates the mechanism (Figure 3). With a rare outcome (prevalence .05) and small samples there is simply not enough signal-bearing data to tamper with: at $N = 20$, 60% of datasets cannot be pushed to significance by dropping at all as the strategy exhausts the table's hostile cells first. Where tampering is structurally possible, rare outcomes are cheap to tip, because each fabricated or relabelled case moves the test statistic disproportionately.

3.2. Strategies that are plausibly hardest to detect are among the cheapest

Strategies in every figure are ordered by their median proportion tampered, computed from the data at render time, running from the most expensive strategy on the left to the cheapest on the right. The forensically critical result is where the label-switching family sits in that ordering. In the two-group design, condition switching is the second cheapest of the five strategies (median 5 switched labels at $N = 100$) despite altering no recorded value, only condition labels, and it succeeds in every dataset at every sample size examined. In the 2×2 setting the corresponding strategy is the cheapest of the three. Its correlation analogue, re-pairing (median 8 re-paired values at $N = 100$), is not the cheapest in its setting. High-leverage replacement and fabrication are cheaper, but it costs about the same as unprincipled case deletion and less than duplication, while preserving both marginal distributions exactly. Neither manipulation alters anything that a check of one variable in isolation could flag; only the joint structure is disturbed. This is, however, a procedural argument for stealth, of the kind asserted by Chambers (2017), rather than a measured one: the statistical detectability of these manipulations by multivariate or forensic tools was not examined here and requires dedicated study.

Re-pairing carries one important qualification that the other strategies do not. It fails outright in 21.8%–30.8% of datasets, at every sample size, and the failures are a property of the greedy rule rather than of the data running out. The procedure gives up when the most discordant case already holds the y-value matching its own x-rank, or when the indicated swap would not increase r; at that point it stops with the correlation positive but short of significance, typically after only a handful of swaps. A tamperer willing to consider swaps beyond the single rank-matched one would not be stopped this way, so this is a limitation of the specific procedure simulated, not evidence that re-pairing is intrinsically unreliable. Taken together, then: efficiency and likely detectability are not in tension for the would-be fraudster, since in each setting at least one of the cheapest strategies is also among the least visible to routine univariate checks — cleanly so in the two-group and 2×2 designs, and in the correlation design for the roughly three-quarters of datasets on which this implementation of re-pairing succeeds at all.

Table 2: Tampering effort at a representative sample size (total N = 100), by setting and strategy. Values in parentheses are +/-1 Monte Carlo standard error. Never is the proportion of datasets for which significance was never attained, tampering having run to a structural limit or having stalled with no improving move available; the latter is what drives the high value for correlation re-pairing. The two budget columns are post hoc tampering budgets: the proportion of datasets needing more than that share of the sample tampered, never-significant datasets exceeding every budget by definition. The median is computed over all iterations with never-significant ones ranked above every finite count, and is also given as a proportion of the sample (Prop.). PI is the 95% prediction interval, the 2.5th-97.5th percentile of the tamper counts across iterations; its bounds are themselves Monte Carlo estimates, so each bound carries its own bootstrap MCSE in parentheses. The figures show interquartile ranges instead, for the reason given in the Methods. D/f/r denotes the duplication, fabrication and replacement strategies, which coincide for binary data.

Setting	Strategy	Never	> 10%	> 25%	Median	95% PI	Prop.
Mean difference	Condition switching	0.000 (0.000)	0.044 (0.002)	0.000 (0.000)	5.0 (0.0)	[1.0 (0.5), 11.0 (0.1)]	0.050 (0.000)
	Dropping	0.000 (0.000)	0.440 (0.005)	0.009 (0.001)	10.0 (0.1)	[1.0 (0.5), 23.0 (0.2)]	0.100 (0.001)
	Duplication	0.000 (0.000)	0.685 (0.005)	0.107 (0.003)	14.0 (0.0)	[2.0 (0.9), 32.0 (0.0)]	0.140 (0.000)
	Fabrication	0.000 (0.000)	0.651 (0.005)	0.016 (0.001)	14.0 (0.0)	[2.0 (0.9), 24.0 (0.0)]	0.140 (0.000)
	Replacement	0.000 (0.000)	0.088 (0.003)	0.004 (0.001)	4.0 (0.0)	[1.0 (0.5), 16.0 (0.4)]	0.040 (0.000)
Correlation	Dropping	0.000 (0.000)	0.370 (0.005)	0.006 (0.001)	9.0 (0.2)	[1.0 (0.3), 21.0 (0.4)]	0.090 (0.002)
	Duplication	0.000 (0.000)	0.662 (0.005)	0.127 (0.003)	14.0 (0.0)	[1.0 (0.3), 37.0 (0.4)]	0.140 (0.000)
	Fabrication	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	4.0 (0.0)	[2.0 (0.6), 6.0 (0.0)]	0.040 (0.000)
	Re-pairing	0.239 (0.004)	0.375 (0.005)	0.239 (0.004)	8.0 (0.1)	[2.0 (0.6), never]	0.080 (0.001)
	Replacement	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	3.0 (0.0)	[1.0 (0.3), 6.0 (0.0)]	0.030 (0.000)
2x2, prev. 0.05	Condition switching	0.006 (0.001)	0.146 (0.004)	0.034 (0.002)	5.0 (0.0)	[1.0 (0.1), 30.0 (0.1)]	0.050 (0.000)
	Dropping	0.076 (0.003)	0.757 (0.004)	0.467 (0.005)	25.0 (0.0)	[1.0 (0.1), never]	0.250 (0.000)
	D/f/r	0.006 (0.001)	0.639 (0.005)	0.044 (0.002)	13.0 (0.0)	[1.0 (0.1), 28.0 (0.4)]	0.130 (0.000)
2x2, prev. 0.2	Condition switching	0.000 (0.000)	0.290 (0.005)	0.000 (0.000)	8.0 (0.5)	[0.0 (0.5), 16.0 (0.5)]	0.080 (0.005)
	Dropping	0.000 (0.000)	0.656 (0.005)	0.061 (0.002)	13.0 (0.2)	[0.0 (0.5), 29.0 (0.1)]	0.130 (0.002)
	D/f/r	0.000 (0.000)	0.815 (0.004)	0.251 (0.004)	19.0 (0.0)	[0.0 (0.5), 38.0 (0.2)]	0.190 (0.000)

4. Discussion

The distance from a null dataset to a statistically significant one is short and cheap. The median effort is a small percentage of the sample across strategies and settings, and at $N = 400$ no strategy in either continuous setting requires more than 10% of the sample in as many as a quarter of datasets. Where tampering fails outright, it fails for one of two reasons, and neither is that the tampering was merely expensive. In the rare-outcome 2×2 cells too few hostile cases exist to move the table at all, so the failure is structural: at prevalence .05 and $N = 20$, 60% of tables cannot be pushed to significance by dropping. In correlation re-pairing the failure is algorithmic instead — the greedy rule stalls while the correlation is still positive but short of significance — and it accounts for the largest block of never-attained results in the study. The share of the sample that must be altered falls as samples grow, by between 6.2- and 10.0-fold across the range examined, so the intuition that a large study is protected by its size is not merely optimistic but inverted. This extends the finding for p-hacking that larger samples alone do not reduce its success rate (Stefan & Schönbrodt, 2023). The strategies that preserve univariate distributions (condition switching and re-pairing) are among the most effective, and are plausibly also the hardest to detect — though that is a procedural argument rather than a tested one, and further research is necessary to understand the relative statistical detectability of each of these strategies.

These results give the opening cases their denominator. In the vasculitis trial, re-adjudicating the remission status of five patients out of 331 (1.5% of the sample) separated a failed trial from a successful one. Nothing there was invented and nothing was deleted: five cases were moved from one cell of a 2×2 table to another, which is the operation simulated here as label switching in the binary design, and the cheapest of the three strategies available in that design (median 3.8% of the sample at $N = 400$ and prevalence .20). Against the distributions estimated here, 1.5% is not an aberration but a typical cost: across the continuous-setting strategies at $N = 400$, the median share of the sample that must be tampered lies between 1.2% and 6.5%.

The same reading applies to the deletion cases, such as the post hoc exclusions documented by Schiller et al. (2018), in which data points were dropped to 'reveal' an effect the researchers appear to have genuinely believed was there. It is worth being precise about what makes such deletion dangerous, because excluding data is not itself misconduct: principled outlier exclusion is specified in advance, justified by evidence that a measurement is invalid (equipment failure, inattention, impossible values), and applied blind to which way the excluded point pushes the estimate. The dropping strategy simulated here is its exact inverse — a case is removed *because of* the direction it pushes the result, with the test re-run after each removal — and that single reversal is what converts a defensible practice into a reliable false-positive generator. At $N = 100$, it takes the median null dataset across the significance threshold by deleting 10.0% of the sample in the two-group design and 9.0% in the correlation design.

Deleting real observations to reveal an effect one believes is genuinely there plausibly feels like a lesser violation of research integrity than inventing observations that were never collected, and the survey evidence indicates that it is treated as one. The simulations give that intuition no support. At the same sample size, fabrication requires 14.0% and 4.0% of the sample in the two settings, both strategies succeed on essentially every dataset

(Table 2), and both yield the same end product: a significant result where no effect exists. The gap between their costs, at most 5.0% of the sample, is a difference of degree rather than the difference in kind implied by their standing in the survey data, where post hoc exclusion is rated 1.61 out of 2 for defensibility and falsification 0.16 (John et al., 2012). In their impact on the false-positive rate the two acts are roughly equally dangerous, and proportionately very little of either is needed. Whatever distinguishes unjustified deletion from fabrication, it is not that deletion is the weaker instrument.

The regulatory definitions classify these acts by mechanism (i.e., was the data point invented, altered, or omitted (Steneck, 2007)), but the clause that unifies them is a consequence criterion: that the research is not accurately represented in the research record. Every strategy simulated here satisfies it identically as each produces a result that did not exist before.

The practical asymmetry, then, is not in the acts but in how they are regarded. Alongside the defensibility ratings above, excluding inconvenient participants is admitted by 38–43% of researchers and falsifying data by under 2% (John et al., 2012). When respondents are given provable deniability, admitted fabrication and falsification converge to indistinguishable rates (Gopalakrishna et al., 2022) — evidence that the gap is substantially one of reporting rather than of behaviour. And since perceived seriousness is the strongest available predictor of engagement (Entradas et al., 2026), a seriousness ranking that is mis-set on the dimension these results measure is not merely an intellectual inconsistency: it is a live cause of the behaviour. Deleting a hostile case and inventing a favourable one are, in effort and in damage to the evidence, the same act.

The ethics of reporting the results of this simulation study were given considerable thought. Although the simulations were designed to make a continuum visible rather than to provide a recipe, in principle, this article documents knowledge that could be weaponised by bad actors. At the same time, the tampering strategies simulated here have been publicly described for years (Brown, 2020; Chambers, 2017); what is added here is quantification. This quantification could help change future scientific behavior, by raising awareness around how impactful even a very small number of data tampers can be.

The strategies employed in this simulation are deliberately simple and myopic, tampering one step at a time with no lookahead. Real, detection-aware fraud could plausibly trade efficiency for plausibility. However, there are of course many cases of detection-unaware fraud caught because of impossible or implausible results and data, and the prevalence of undetected, relatively expert research fraud is unknown by its very nature. Conversely, several naive strategies here leave potentially detectable signatures (e.g., truncated distributions after dropping, leverage outliers after fabrication, or shifted marginal prevalence in the 2×2). Future research is needed to examine such detectability via statistical means and modern forensic tools (Brown & Heathers, 2017; Heathers et al., 2018; Wilkinson et al., 2025). However, the distribution-preserving strategies imply that likely statistically invisible tampering is cheap and effective. Auditable chain-of-custody for data, such as those employed in medical clinical trials that allowed the US FDA to detect abnormalities in the trial discussed in the introduction, would be needed to prevent or detect such forms of tampering (U.S. Food and Drug Administration, Center for Drug Evaluation and Research, 2026). Infrastructure for auditable chain-of-custody is rare in other fields.

Several further limitations should be noted. First, the fabrication and replacement strategies require a choice of how extreme the invented values are, and that choice is essentially arbitrary: once an analyst is willing to make up data, values of any extremity are available, so the simulated values occupy one point in a large space of experimenter degrees of freedom trading efficiency per fabricated point against plausibility. The values used here are deliberately aggressive (e.g., intervention scores drawn from a distribution five standard deviations above the truth), so the relative cost of these strategies is harder to benchmark than that of strategies that only delete or relabel existing data, and their position in the effort ordering should be read as a property of these particular parameter choices rather than of fabrication as such. Second, all data-generating processes here are exact nulls. Arguably the more common real-world scenario is a small true effect "topped up" to find statistical significance, which would serve to inflate effect sizes. The effort required should fall steeply with the true effect size, and simulating non-null data-generating processes would both quantify that and connect these results more directly to the questionable-research-practices literature, where a real but overestimated effect is the typical outcome. Third, the simulation design was not preregistered. This project began in 2022, before the publication of the standardized planning and preregistration template for simulation studies in psychology (Siepe et al., 2024), and both the design and my approach to specifying simulation studies matured considerably over that period; the ADEMP specification reported here therefore documents the final design transparently rather than constraining it in advance.

Data and code availability

All simulation code and results, including extended ADEMP specifications, per-condition tables, and Monte Carlo standard errors for every estimate, are available at https://github.com/ianhussey/simulation-impact-of-data-tampering.

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